Electromagnetic Characterization of a Radially-Slotted Circular Microstrip Patch Antenna in the C-Band

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***Abstract*—This paper presents the design and electromagnetic characterization of a circular microstrip antenna exhibiting multiband operation in the 5–7 GHz frequency range. The radiating circular patch is modified by etching out eight tilted rectangles, positioned at an optimized radial distance from the patch center to introduce multiple resonant modes. To enhance radiation performance, a novel ground-plane configuration is employed in which a rotated replica of the top radiating patch is used instead of a conventional full ground plane. This configuration strengthens the fringing fields and promotes enhance the radiation from the etched-out portions. The antenna is excited using a coaxial probe feed. Full-wave simulations shows three distinct resonant bands centered at 5.55 GHz, 5.80 GHz, and 6.40 GHz. The dominant resonance at 5.80 GHz exhibits a return loss of 25.68 dB and a peak gain of approximately 8.8 dB. The proposed design achieves stable multiband behavior without a significant increase in antenna size, which makes it suitable for compact C-band and sub-6 GHz wireless applications.**

***Index Terms*—C-band, circular patch antenna, radial slots, multiband antenna.**

I. Introduction and Literature Review

Microstrip patch antennas still hold the pivotal position in the design of compact wireless systems due to their easy fabrication, low profile, and compatibility with printed circuit boards (PCBs). Among the available configurations, circular microstrip patches provide stable radiation characteristics with a small footprint and thus are highly compatible with C-band and sub-6 GHz applications, such as short-range sensing, WLAN communication, and emerging wireless devices [1]–[5]. Despite this, a single conventional circular patch antenna can usually only support one dominant resonant mode and has a relatively narrow impedance bandwidth. Therefore basic configurations are often insufficient to meet the performance requirements of modern wireless communication systems.

Researchers explored modifications to the surface of circular and rectangular microstrip patches by selectively etching out portions to deliberately disturb the surface current density. Numerous studies have demonstrated that the use of certain slot geometries, including rectangular slots and H-shaped cuts, is an efficient way to disturb the current paths and make it possible to excite additional resonant modes without a major enlargement of the antenna [2]–[4]. The research on multiple slot configurations has also revealed better control of interband frequency spacing as well as increased tuning flexibility of multiband antenna responses.

Another factor that influences the performance of slot-loaded microstrip patches is feeding configuration. Even small variations in feed position can affect the input impedance significantly. Studies on coaxial probe-fed circular patch antennas have confirmed this sensitivity. They are useful mainly when slot discontinuities are located close to the feed point, leading to shifts in impedance matching and resonant frequencies [5]. Although alternative feeding techniques such as microstrip line and coplanar waveguide (CPW) feeds are available, the coaxial probe feed remains a practical and widely adopted choice for simple antenna prototypes due to its simplicity and ease of implementation.

Modifications to the ground plane, such as defected ground structures (DGS), play a significant role in designing better radiation characteristics of microstrip antennas. The introduction of such discontinuities divide surface wave currents, which in turn may lead to improvements in impedance bandwidth and radiation efficiency. DGS- and artificial magnetic conductor (AMC)-based patch antennas have been reported to achieve stable multiband performance in the sub 6 GHz frequency range [6], [7]. Slot-based perturbations combined with ground-plane modifications have also been used to realize circularly polarized antennas for navigation, satellite, and 5G applications [8]–[10]. In addition, slot-loading techniques have been extended to substrate-integrated waveguide (SIW) structures, where slot placement and cavity excitation strongly influence the electromagnetic response [11].

Recently, comparative studies focusing on the influence of slot geometry on surface current behavior have improved our understanding of slot-loaded circular patch antennas. In [12], the effects of circular and square slot configurations on a circular patch antenna were investigated well, showing that slot shape plays a major role in determining resonant modes,

impedance-characteristics, and radiation behavior. Additionally, recent DGS-based circularly polarized patch antenna designs reported in [13], [14] demonstrate how ground plane discontinuities and unconventional radiating structures can be used to enhance bandwidth and radiation performance for modern 5G and sub-6 GHz systems

Inspired by these research efforts, the present work focuses on the electromagnetic characterization of a radially slotted circular microstrip patch antenna operating in the C-band. Uniformly distributed tilted rectangular radial slots are introduced on the radiating circular patch. The ground plane is patterned using the same slot configuration as that of the top face but rotated by a fixed angle. Rather than targeting polarization diversity or wideband operation, this study aims to analyze how radial slot loading and ground-plane rotation influence impedance behavior, resonant mode generation, and radiation characteristics in a compact circular patch antenna operating within the 5–7 GHz frequency range. The design process and the antenna characterization are discussed in detail in the upcoming sections.

II. Antenna Design and Simulation Methodology

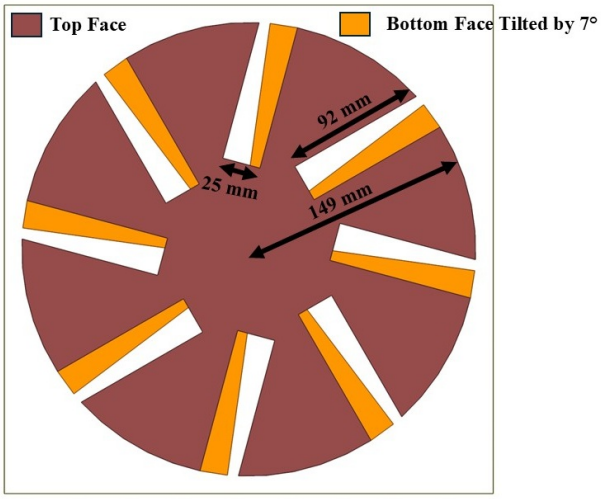


Fig. 1: Design of the antenna.

The antenna is implemented on a Rogers RT/duroid 5880 substrate of thickness 1.57 mm and relative permittivity of 2.2. The radiating element consists of a circular patch into which eight tilted rectangular slots are cut radially. As illustrated in Fig. 1, these slot cuts interrupt the natural radially outward current flow of the patch in a controlled way. The total radius of the circular section is 149 mm, and the rectangles are of length 92 mm and breadth 25 mm. Rectangles are tilted by an angle of 75° which creates additional resonant paths without increasing the overall size of the structure by much. To support and shape the behavior of the slot-loaded patch, the ground plane is also patterned using the same geometry of the top face but rotated by a fixed angle of 7°, as shown in Fig. 1. Positioning the ground-plane discontinuities beneath the slot openings on the top surface influences the electric field distribution and helps improve impedance matching characteristics across the operating bands. A coaxial probe feed at the geometric center is used to excite the antenna.

The structure is analysed using Ansys HFSS. A driven-modal setup is used with a frequency sweep covering 5–7 GHz. Radiation boundaries are placed sufficiently far from the antenna to approximate free-space conditions. The simulation evaluates the reflection coefficient, realized gain, radiation patterns, and surface current distributions. The combined influence of the slot geometry, rotated ground plane, and coaxial probe feed results in three distinct resonant modes, consistent with the measured behavior of the prototype.

III. Results and Analysis

This section presents the simulated electromagnetic performance of the radially-slotted circular microstrip patch antenna. The analysis focuses on the return loss, radiation characteristics, and polarization behavior in order to examine the influence of rectangular slot on the antenna performance.

A. Return Loss Performance

The simulated return loss plot of the proposed antenna is illustrated in Fig 2. Good impedance matching with the 50 coaxial port is observed at 5.8 GHz as the return loss goes to approximately 25.6 dB. Its operating frequency ranges from 5.77 GHz to 5.84 GHz, which corresponds to a usable bandwidth of about 70 MHz within the C band. The emergence of extra resonant behavior in the response is, to a large extent, the explanation of the perturbation of surface currents caused by the rectangular radial slots as shown in Fig 3. These slots break the normal flow of the circumferential current of the circular patch and make the currents to follow the longer way along the slot edges.

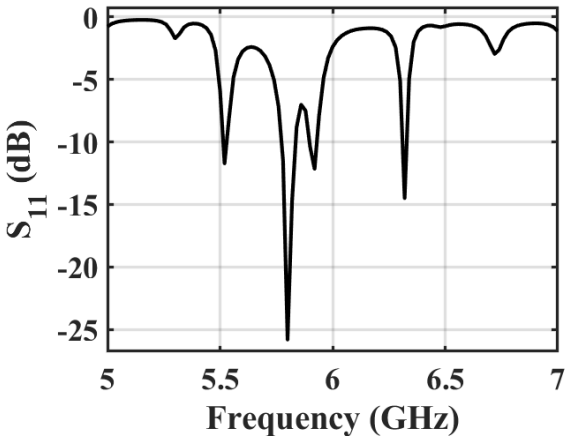


Fig. 2: Simulated S_{11} response of the antenna.

The simulated surface current distribution as shown in Fig. 3, confirms that strong currents are concentrated along the edges of the rectangular radial slots, particularly near the slot terminations. In a conventional circular microstrip patch

antenna, the surface currents predominantly flow along the patch periphery, corresponding to the dominant TM_{11} mode. In contrast, in the proposed design, we introduced radial slots which disrupts this natural current flow. They force the currents to follow elongated paths along the slot edges. This perturbation increases the electrical length of the radiating structure and helps in excitation of additional resonant modes. As a result, the antenna exhibits a multi resonant S_{11} response along with modified radiation characteristics.

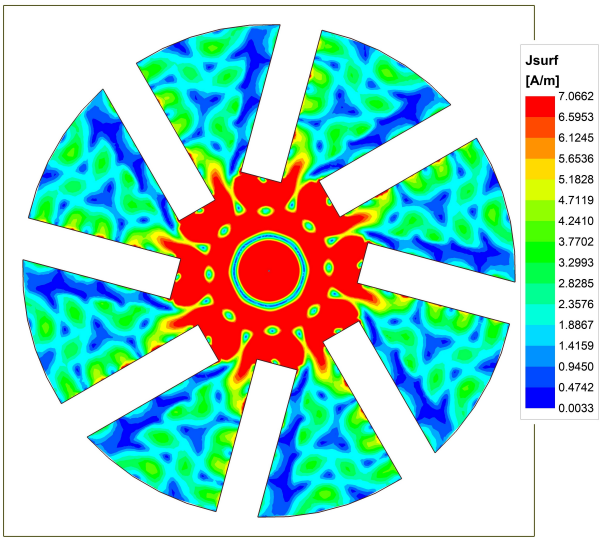
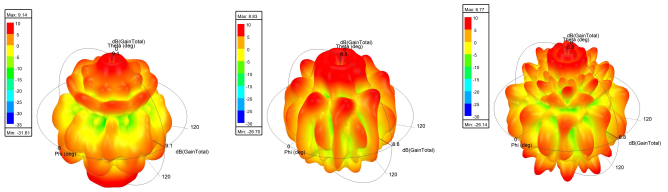


Fig. 3: Surface current distribution of the antenna.

B. Gain Characteristics

The simulated three dimensional radiation patterns of the antenna in Fig. 4 shows antenna at the three operating frequencies. The antenna has a quasi-omnidirectional radiating characteristic in the azimuth plane, and there are a number of radiation lobes observed in the elevation plane. The main operating is at 5.8 GHz frequency, where the antenna has a peak realized gain of 8.8 dB. The largest gain among the three that was realized. bands is captured at a frequency of 5.52 GHz at about 9.1 dB, Although, a gain of approximately 6.8 dB is realized at 6.32 GHz. We can observe that there are more than one lobes in the elevation plane, which we can observe in 3D pattern and 2D pattern. Those are the result of excitation of slot loaded higher order resonant modes, which are present in patch geometry. These modes are helpful in bringing about an overall improvement in the achieved gain. They also give additional sidelobes in the elevation plane. Such sidelobe formation is a trade-off between enhancement of gain enhancement and pattern smoothness which is common in multiresonant slot based. patch antennas. However the total radiation characteristics can still be used with small C-band and less than 6 GHz where moderate level of sidelobe is tolerable.



(a) Gain at 5.52 GHz (b) Gain at 5.8 GHz (c) Gain at 6.32 GHz

Fig. 4: Simulated gain patterns at three resonant frequencies.

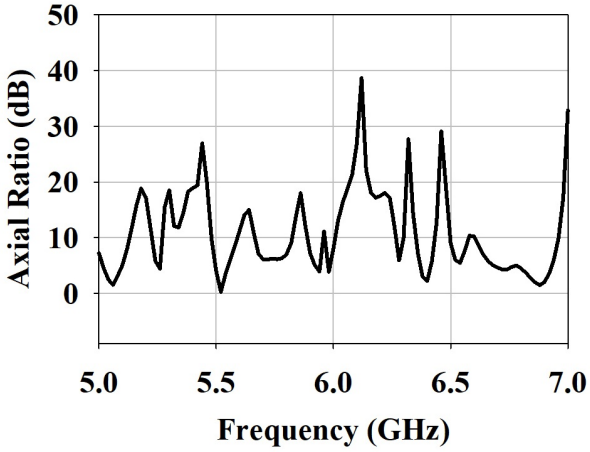


Fig. 5: Simulated axial ratio response of the antenna.

C. Axial Ratio Analysis

The simulated axial ratio response of the antenna as a function of frequency is shown in Fig. 5. At the dominant operating frequency of 5.8 GHz, the axial ratio remains well above the 3 dB threshold, indicating that the antenna radiates predominantly linearly polarized waves at this frequency. The absence of an axial ratio dip below 3 dB confirms that circular polarization is not supported within the operating band. This behavior is consistent with the antenna geometry and excitation mechanism, where the slot-loaded circular patch and coaxial probe feed primarily excite a single dominant field component without the required orthogonal modes in quadrature phase necessary for circular polarization.

Although the introduction of radial slots significantly perturbs the surface current distribution to enable multiband operation and gain enhancement, it does not introduce the modal degeneracy or phase orthogonality required to achieve circular polarization. Consequently, the observed axial ratio response is characteristic of a linearly polarized radiator, which is suitable for C-band and sub-6 GHz wireless applications where polarization diversity or circular polarization is not a primary requirement.

D. Parametric Analysis

Based on the parametric analysis shown in Fig. 6 and Fig. 7, the optimal design parameters are selected by examining their impact on impedance matching and the stability of the

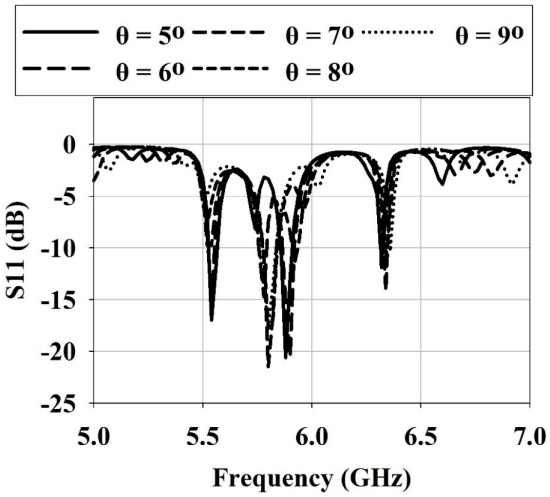


Fig. 6: Parametric Sweep of the Slot Angle (θ).

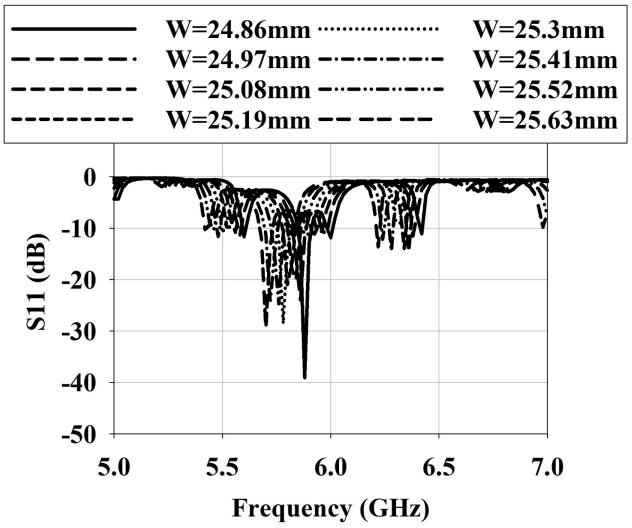


Fig. 7: Parametric Sweep of the Slot Width

resonant modes. Fig. 6 demonstrates the impact of ground-plane rotation angle. It can be seen that using smaller rotation angles leads to the patch and ground structure having lesser coupling, which means poor impedance matching, while larger angles bring about distortion in the resonance behavior. The rotation in the range of approximately 7 is optimal and it results in achieving a good return loss and multiband features that are very stable and thus chosen.

Fig. 7 illustrates how the reflection coefficient is affected by the slot width (W). It is clear that slot width plays a major role in changing the current distribution and coupling inside the patch. Narrower widths result in insufficient perturbation; on the other hand, wider widths lead to worsening of impedance matching and also to shifting of resonant frequencies. A width of about 25.3 mm gives the deepest resonance and also better matching, so it is selected. Therefore, the selected parameters represent an optimal balance between coupling, impedance matching, and resonance stability, ensuring the desired multiband performance of the proposed antenna.

Table I presents a comparison of the proposed antenna with recently reported slot-loaded and defected ground structure (DGS)-based microstrip antennas. It is found that the existing designs have been aimed at the realization of broad impedance band or small size, however, they are typically of low to moderate gain factors. In comparison, the proposed antenna has shown a definite enhancement in the radiation performance as it has a peak realized gain of 8.8 dBi and multiband operation at frequencies of 5.55 GHz, 5.80 GHz and 6.40 GHz. This is primarily because of the addition of radially arranged tilted slots which are effective in increasing current path and creating several resonant modes and the rotated ground-plane configuration which enhances interaction of fields and radiation efficiency. Though the bandwidth is quite small, the design is optimized to be used specifically in C-band, and the band provides stable and efficient operation. The proposed antenna is overall a superior compromise in terms of gain, compactness and multiband capability to the reported works.

IV. Conclusion

A compact circular microstrip patch antenna employing uniformly spaced tilted rectangular radial slots as the top and bottom plane is investigated for multiband operation in the 5–7 GHz frequency range. The introduction of radial slots perturbs the natural surface current distribution of the circular patch, creating multiple effective resonant paths without increasing the antenna size. As a result, three distinct resonances are obtained at approximately 5.55 GHz, 5.80 GHz, and 6.40 GHz, with the dominant resonance at 5.80 GHz exhibiting strong impedance matching and a minimum reflection coefficient of about -25.68 dB. At this frequency, the antenna achieves a peak realized gain of nearly 8.7 dBi. The radiation characteristics show a quasi-uniform field distribution in the azimuth plane due to the combined effect of coaxial probe excitation and the rotational symmetry of the radial slot arrangement, while multiple elevation-plane lobes are observed as a consequence of higher-order modes excited by slot-induced current perturbations. The axial ratio response confirms that the antenna radiates linearly polarized waves, consistent with the single-feed configuration and the absence of orthogonal mode degeneracy. Overall, the results demonstrate that the combined use of radial slot loading and a rotated slotted ground plane provides an effective and compact solution for achieving stable multiband behavior and enhanced radiation performance in C-band and sub-6 GHz wireless applications.

The antenna proposed can be used in the C-band and sub-6 GHz wireless applications that would include short-range high-frequency communication systems and small wireless

TABLE I: Comparison with Existing Antennas

Ref	Frequency (GHz)	Technique	Bandwidth	Gain (dBi)
[3]	3.32–5.96	H-shaped slot with partial ground plane	2.64 GHz	2–3
[4]	2.75–4.23	Wrench-shaped slot with parasitic elements	1.48 GHz	2.77
[6]	1.17–2.49	Slotted patch with defected ground structure (DGS)	Single-band	2–3
[13]	3.07–4.47	Multi-slot antenna with DGS for circular polarization	$\sim$1.3 GHz	> 6
This Work	**5.55–6.40**	**Radial slots with rotated ground plane**	**70 MHz**	**8.8**

modules where higher gain is necessary to enhance the reliability of the links. The gain of about 8.8 dBi is desirable in the design to suit applications in the point-to-point links and localized communication systems in the frequency range of 5-7 GHz. But there are also some limitations of the antenna. Its bandwidth of impedance is relatively low (around 70 MHz), and hence, it can be used only in limited broadband communication systems. Also, higher-order modes are excited in the radiation pattern as multiple lobes are visible in the elevation plane, and this is not desired in applications where smooth or strongly directional radiation properties are needed. In addition, the antenna has a linear polarization, and this limits its application in a system that needs polarization diversity or circular polarization. Irrespective of these constraints, the proposed design is a viable trade-off between gain improvement, small design and multiband performance. This makes it appropriate to particular C-band applications in which high gain and constant multi-resonant performance are more important than wideband performance.

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